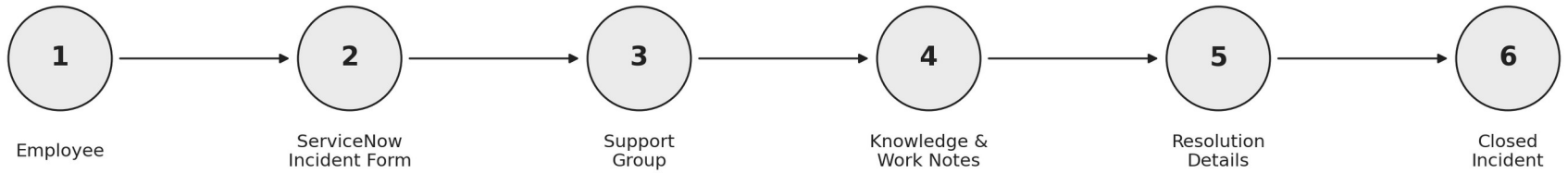
Project Design Phase-II
Data Flow Diagram & User Stories

Date	21 October 2026
Team ID	
Project Name	Incident Lifecycle Automation in ServiceNow
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a visual representation of how information moves through a system. It shows who or what sends data into the system, which processes change that data, where the data is stored and what is sent back out. For this project, the DFD shows how an incident moves through ServiceNow from the moment it is reported until it is closed.

Example: (Simplified)

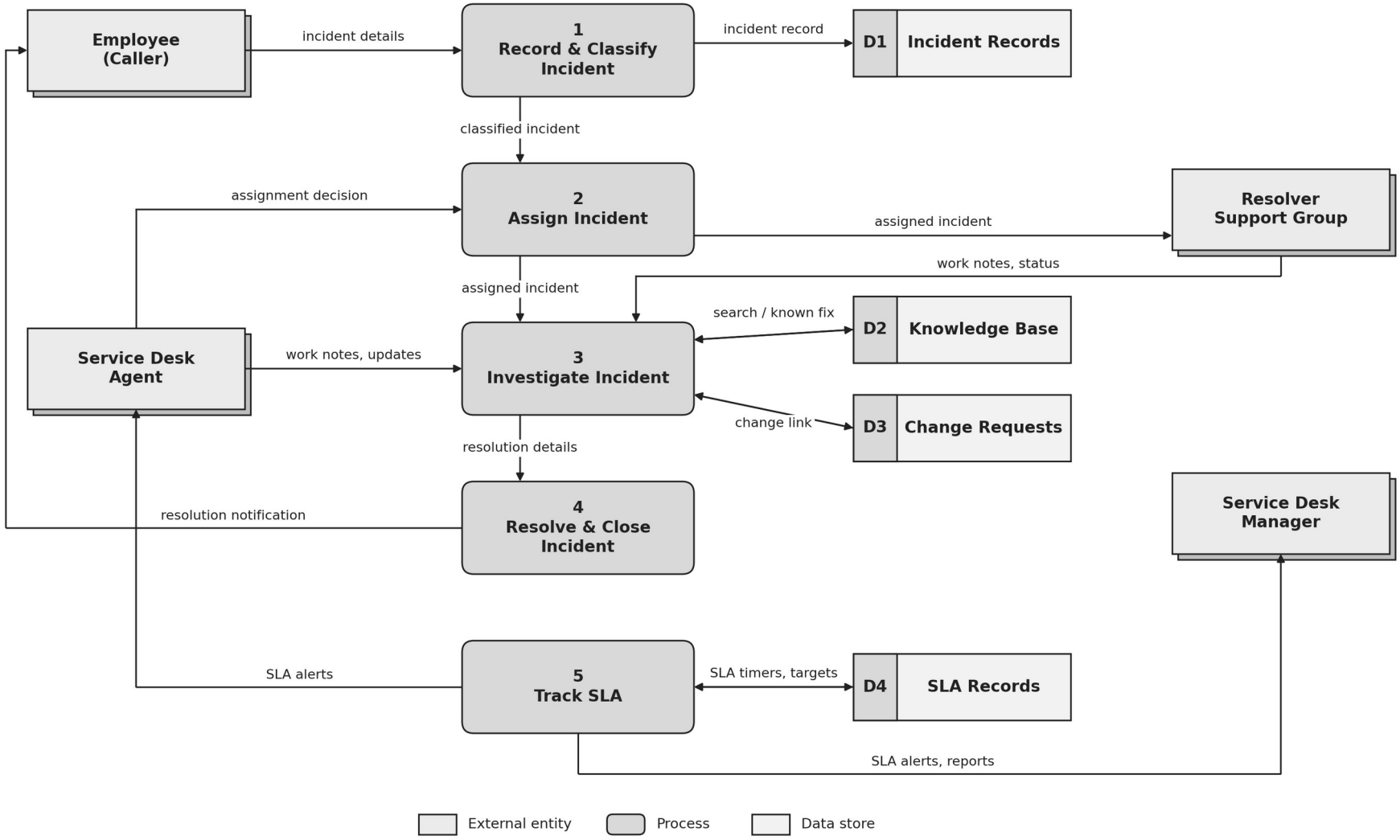


1. The employee reports an IT issue and the service desk agent records it in ServiceNow.
2. The agent selects the category, urgency, service and configuration item, and assigns the incident to a support group.
3. The support group investigates the incident using knowledge articles, related incidents and work notes.
4. A change request is linked if a change is needed to fix the issue.
5. The cause, workaround and resolution notes are recorded and SLA status is tracked.
6. The incident is closed and the employee is notified.

DFD Components:

External entities	Employee (Caller), Service Desk Agent, Resolver Support Group and Service Desk Manager
Processes	1 Record & Classify Incident, 2 Assign Incident, 3 Investigate Incident, 4 Resolve & Close Incident, 5 Track SLA
Data stores	D1 Incident Records, D2 Knowledge Base, D3 Change Requests, D4 SLA Records

DFD Level 0: Incident Lifecycle Automation in ServiceNow



User Stories

The table below lists the user stories for the incident lifecycle, grouped by user type and functional requirement (epic).

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Employee (Caller)	Incident Recording	USN-1	As an employee, I can report an IT issue by describing the problem and my contact details.	The incident is created and I receive an incident number.	High	Sprint-1
		USN-2	As an employee, I receive a confirmation once my incident is recorded.	I receive a notification with the incident number and short description.	Medium	Sprint-1
	Incident Tracking	USN-3	As an employee, I can check the status of my incident.	I can see the current state and the assigned support group.	Medium	Sprint-2
		USN-4	As an employee, I am notified when my incident is resolved.	I receive a message with the resolution summary.	High	Sprint-2
Service Desk Agent	Classification	USN-5	As an agent, I can set the category, subcategory, urgency, service and configuration item on an incident.	The incident saves with all classification fields filled in and visible.	High	Sprint-1
		USN-6	As an agent, I can set the impact and urgency so that the priority is calculated.	The priority updates according to the selected impact and urgency.	High	Sprint-1
	Assignment	USN-7	As an agent, I can assign an incident to the correct support group.	The incident shows the assignment group and appears in that group’s queue.	High	Sprint-1
		USN-8	As an agent, I can reassign an incident when it was routed to the wrong group.	The reassignment is saved with a note and the new group can see the incident.	Medium	Sprint-2
	Investigation	USN-9	As an agent, I can search knowledge articles from the incident.	Matching articles are listed and I can attach one to the incident.	Medium	Sprint-2
		USN-10	As an agent, I can add work notes to record the troubleshooting steps.	Notes are saved with the time and author and shown in the activity history.	High	Sprint-1
		USN-11	As an agent, I can link related incidents to a parent incident.	Child incidents show the parent and the parent shows its child incidents.	Medium	Sprint-2
Support Group Member	Resolution	USN-12	As a support group member, I can see the incidents assigned to my group.	The group queue lists the incidents with their priority and state.	High	Sprint-1
		USN-13	As a support group member, I can update the state and add work notes while investigating.	The state and notes are saved and visible to the agent.	High	Sprint-2

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
		USN-14	As a support group member, I can record the cause, workaround and resolution notes before resolving.	The incident cannot be resolved until the resolution details are entered.	High	Sprint-2
	Change Handling	USN-15	As a support group member, I can create or link a change request when a change is needed to fix the incident.	The change request appears in the related records of the incident.	Medium	Sprint-3
Service Desk Manager	SLA Monitoring	USN-16	As a manager, I can view the SLA status of open incidents.	SLA progress and breach status are shown for each incident.	High	Sprint-2
		USN-17	As a manager, I am alerted when an SLA is close to breaching.	An alert is sent when the SLA passes its warning point.	Medium	Sprint-3
	Reporting	USN-18	As a manager, I can view reports of incidents by category, priority and state.	The report shows incident counts and can be filtered.	Low	Sprint-3
Administrator	Configuration	USN-19	As an administrator, I can configure categories, assignment groups and SLA definitions.	The changes are saved and applied to new incidents.	High	Sprint-1
		USN-20	As an administrator, I can manage user roles and access.	Users only see the features allowed by their role.	Medium	Sprint-1

Priority and Release Notes:

High priority stories are needed for the core incident lifecycle (recording, classification, assignment, investigation and resolution). Medium priority stories improve efficiency and visibility. Low priority stories add reporting and can follow later.

Sprint-1 covers the core lifecycle and configuration, Sprint-2 adds tracking, reuse of knowledge and related incidents, and Sprint-3 adds change handling, SLA alerts and reports.